

Engines and Individuals

Token conversion between composed learners, the cycle space as a price code, and error correction as the criterion of individuation

F1R3FLY.io Research Notes

2026-08-01

Abstract

The companion note [3] composes learners — populations of mortal scientists together with their resource reservoirs — by parallel composition, and reads the ecological relation between two learners off a grading vector over four typed namespaces. That note works in a single token colour. Here we give each community its own colour and adjoin *engines*: conversion contracts of the kind the virtual-token note [7] prices, realised with the financial combinators of [11] in the cost-accounted form of [8]. Three things follow.

First, the engine is not a contract but an organism. It is a fourth row in the access-profile table of [1, §10] — persistent and non-exclusive like a source, exhaustible per colour like prey, and alone among the four in being *adaptive in its rate*. Its dissipation θ , posited as a parameter in [7], is here endogenous: it is the metering charge of the rendezvous that implements the conversion. The combinators become its reaction language, with one extension — the observable that sets the rate reads the issuer’s own inventory, so arbitrage-freedom becomes a property of a *marking* rather than of a system.

Second, the engine graph carries an error-correcting code on prices, and we can say exactly how strong it is. The syndrome of a corrupted rate has squared norm $a^2(1 - R_{\text{eff}}(e))$, where R_{eff} is effective resistance; by Foster’s theorem the total detection strength summed over engines is exactly β_1 , the first Betti number. A bridge has $R_{\text{eff}} = 1$ and carries no pricing discipline at all. Prices are self-correcting exactly to the extent that the ecology is densely connected.

Third, distance is not a parameter of this calculus but a *term*. Rho’s perfect communication is without loss of generality because a pathway is an interposed process, $\text{for}(y \leftarrow x_1)P \mid C(x_1, x_2) \mid x_2!(Q)$, which may drop or corrupt. The dissipation θ is then the failure of interposition to preserve metering, a dropped token is stranded rather than destroyed — giving the second law a mechanism and exergy a measure — and a corrupted quote is exactly the single-edge fault the price code detects. Media are processes, so they are nameable, so they form a namespace one level up the reflection tower; and the medium namespace of one level is the endpoint namespace of the next.

Fourth, and this is the note’s answer to a question the companion notes leave open, *error correction is what individuates*. Transposing the crossover of [4] onto the cost-weighted engine graph, with name collision as the error channel and re-rooting as repair, a region may close over its namespace — merge, in the sense of [3, §8] — iff its isoperimetric ratio clears a threshold set by $\varepsilon\theta/\rho$. Polynomial growth then gives a finite critical radius and a hierarchy of nested individuals; expansion gives individuals exponentially larger for the same parameters. The structural form of the same criterion is partial closure: an individual owns its internal media and does not own its boundary media, since retransmission requires the power to name what failed. Total closure would be fatal, and internal complexity is not merely permitted but purchased — integration depth is bought with reliability, and reliability is available only on a medium one can repair.

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1 Introduction

1.1 Why the unit moves again

The series this note belongs to has moved its unit of analysis twice. In [1] the unit is a single mortal scientist: a cost-accounted rho computation that forms hypotheses in a namespace logic, pays for its assays, and dies when its stack runs out. In [3] the unit is a learner: a population of such computations together with their reservoirs, serving as a namespace, with a distribution over it, composed with other learners by parallel composition. That note ends by asking whether the series individual/lineage/composite continues.

It does, and the reason it does is that composition of learners yields a learner, so the construction is closed under an associative operator. But closure alone gives arbitrary nesting and no scale structure: a soup of composites has no distinguished levels. Something must pick out *skill*, *organism*, *pod*, *species*, *biosphere* as rungs rather than as arbitrary cuts, and the companion notes do not say what.

Two things were also missing from the economics. [3] works in a single token colour, so its conservation law is the flat statement that a fixed supply Θ is never minted. Real coordination between communities is not a transfer of one fungible fuel; it is *conversion*, at rates, between resources that are not interchangeable — and the arbitrage machinery of [7] exists precisely to handle that. And [3] has no geometry: its composition operator is indifferent to whether the two learners are adjacent or a galaxy apart, though nothing in the framework says the cost of reaching across should be the same in both cases.

This note supplies all three, and they turn out to be one thing.

1.2 The move, stated once

Give each community its own token colour. Adjoin engines between colours and weight each engine by its dissipation θ . The resulting weighted graph is the geometry the framework was missing; its cycle space is a code on prices; and the isoperimetry of that graph is the criterion for when a region may stop trading and start sharing — that is, for when a composition of learners becomes an individual.

1.3 What we claim, and what we do not

We claim: that the engine is derivable as a fourth access profile rather than importable as a new kind of thing (§4); that interposition supplies an error model natively, with θ as its metering defect and stranding as its irreversibility (§5); that the detection strength of a price is $1 - R_{\text{eff}}$ and the total

is β_1 (§6, Theorem 6.6); that inventory-holding engines make arbitrage a property of a marking (§3); that merger is available exactly inside an isoperimetric threshold (§8, Proposition 8.4), with partial closure of the medium namespace as its structural form (Definition 5.12); and that a finite critical radius, together with the tower of media, forces the hierarchy that [3] could only permit (§9).

We do not claim to have derived the rates. [7] takes them as given and so, in the main, do we; §12 says what it would take to derive them and why the numbers of [2] are the natural candidate. We do not claim the error channel has been identified: §8.6 inherits, unresolved, the “which ε is binding” question of [4]. We do not claim the correspondence with observed biology is more than suggestive, though §6.4 notes that the ecological pyramid falls out of Proposition 2 of [7] with no adjustment, which is a check we did not arrange.

2 What is imported

From [1]: the cut and the three grades of access (perception, interaction, reflection); the assay via the **where** clause; the conservation law $\sigma + \kappa = \sigma_0 = \Theta$; the access profile distinguishing a *source* (persistent emitter, rate-limited, non-exclusive, non-adaptive) from *prey* (linear send, exhaustible, exclusive, adaptive); the relaxation lattice and its ultrametric; and the freshness-by-quotation proposition, that for any context C with $C[P] \neq P$ the name $@C[P]$ does not occur in P .

From [2]: the two routes to any predicate — *read* it, priced by formula depth, if the environment has computed it into legible structure, or *drive* it, priced by rendezvous — and the observation that the interesting question is not whether the work is done but who is billed for it.

From [3]: the learner as three layers (population, namespace, distribution) with the distribution *inside*; composition as parallel composition; the factorisation theorem, that namespace disjointness is equivalent to absence of a spanning redex and to $\mu_{12} = \mu_1 \otimes \mu_2$; the grading vector $\mathbf{k} = (k_{\text{met}}, k_{\text{src}}, k_{\text{prb}}, k_{\text{mat}})$ over the four typed namespaces **Met**, **Src**, **Prb**, **Mat**; merger $\bowtie$ as a separate operator; the instability of non-interference and its repair by rooting; and competitive dilution in closed form.

From [5] and [6]: signatures as names, token stacks as first-class terms, wrapping by construction, the cost monad (Cost, η, μ) on ciGSLT and its non-idempotence, and located resource stacks.

From [8]: the correction that a stack floating in the associative-commutative soup is ambient authority, repaired by holding every stack on a channel; the acceptance gate; the netting result; and the four cost-accounting patterns.

From [7]: conversion systems, engines, rates with $r(a \rightarrow b) r(b \rightarrow a) = 1$; the log-rate cochain ℓ ; the equivalence of arbitrage-freedom, exactness of ℓ , and existence of a valuation ν ; the graph-Hodge splitting into energy and arbitrage components; $\dim \ker \delta^* = \beta_1$; arbitrage-removal as an idempotent reflection; the first law (conservation of ν in the lossless regime); the second (dissipation θ makes ν a Lyapunov function); and the insistence that ν is a unit of account and never a medium of exchange.

From [4]: the crossover between a shared substrate and autonomous replication, at critical radius $r^* \sim \sqrt{\rho v / \varepsilon}$ in an error rate ε , a repair throughput ρ , and a communication speed v ; and the closing conjecture that this crossover and the factoring of the OSLF-generated logic over a population are two views of one threshold. That conjecture is what §8 cashes.

From [9]: the persistent boundary-forwarder lemma, which is what makes an encoding compositional and which supplies Proposition 5.1; and the diagnosis, made there, that a mediating process written with a linear receipt models one passage rather than a passage-way.

From [11]: the combinators `zero`, `one`, `give`, `and`, `or`, `scale`, `truncate`, `get`, and observables.

3 Colours, engines, and a fork in the conservation law

3.1 Communities are colours

Definition 3.1 (Coloured learner). A *coloured learner* is a learner $\mathcal{L}$ in the sense of [3, §3] together with a distinguished signature $c(\mathcal{L}) \in \Sigma$, its *colour*: the signature at the head of every stack held at a name in $\text{Met}(\mathcal{L})$.

The colour is not new apparatus. [5] already replaces one universal fuel by a family of signature-indexed resources, one per economically empowered actor; [5, §6] calls the collapse to a single s_0 the monochromatic limit. [3] works in that limit. Taking a community’s colour to be its own signature is the natural un-collapsing, and it is what makes “Terran food and alt-Earth food may not be mutually nutritive” a statement the calculus can express: two learners whose `Met` stacks carry incomparable signatures cannot eat each other’s tokens, however adjacent they are.

Definition 3.2 (Engine). An *engine* between colours a and b is a persistent process holding two located reserves, one of each colour, and offering a linear swap: it accepts one a -token and releases $r(a \rightarrow b)$ b -tokens from its own inventory. Its *toll* is the metering charge of the rendezvous that performs the swap.

3.2 The fork

There is a conservation collision here that must be settled before anything else, because the two notes being joined conserve different things. [3] conserves tokens *per colour*: Θ is fixed and never minted. [7] conserves the scalar ν : an engine consumes one a and releases r units of b , which is not per-colour conserving at all. One must give.

- (i) **Engines hold inventory.** A swap is against stock, so per-colour totals *are* conserved and r is the ratio at which inventory is exchanged. The cost: engines deplete, r moves with the marking, and ℓ becomes a function of state.
- (ii) **Conserve only ν .** Colours interconvert at fixed rates and Θ_ν is the invariant. This matches [7] exactly but abandons the per-signature ledger the scientist note runs on.

We take (i). It preserves the more specific law, it keeps engines mortal (an engine that runs out of one colour stops being an engine, which is what we want of an organism), and its cost — state-dependence — is where the content is.

Proposition 3.3 (Arbitrage is a property of a marking). *Under (i) the log-rate cochain is a function $\ell(M)$ of the marking M of the engine reserves. Consequently the Hodge splitting $\ell = \ell_{\text{en}} + \ell_{\text{arb}}$ of [7, §4] must be taken pointwise in M , and arbitrage-freedom is a condition on a state of the ecology rather than on the ecology.*

Proof. Immediate from Definition 3.2: the rate is computed from the reserves, and a swap changes the reserves. Appendix B, block (4), exhibits a triangle of constant-product engines whose cycle yield is exactly 1 at the balanced marking and departs monotonically as one engine is drained; six trades of ten units carry $\log \ell$ from 0 to -1.003 . $\square$

	persistent	exclusive	exhaustible	adaptive
Source	yes	no	in aggregate	no
Prey	no	yes	yes	yes
Scientist	—as prey, plus grade-3 exposure—			
Engine	yes	no	per colour	in rate

Table 1: The engine as a fourth access profile. It is source-like in persistence and non-exclusivity, prey-like in exhaustibility, and unique in that what adapts is not its behaviour under attack but its *terms*.

Remark 3.4 (Trading is what removes arbitrage). Proposition 3.3 is not a defect. In [7] the projection Π onto the exact cochains is a least-squares correction performed by nobody in particular; the note flags that the operational content of “shaking the arbitrage out” is left abstract. Under (i) it is concrete and it is metabolic: an arbitrageur is a computation that traces a cycle of yield > 1 , and each pass moves the reserves in the direction that reduces the yield. Arbitrage-removal is not applied to the system, it is *eaten out* of the system, and the eating is metered like anything else. This is the sense in which the idempotent reflection of [7, Prop. 3.4] is realised rather than postulated.

Corollary 3.5 (Residual arbitrage). *Since each pass of a cycle costs the arbitrageur at least one token, and the yield per pass falls monotonically toward 1, there is a level of arbitrage below which removal is unprofitable. A mortal ecology therefore carries a floor of arbitrage set by its own metering, and the floor is higher the poorer the ecology.*

Corollary 3.5 is [1, Cor. 19] — poverty is confining — one level up and in a second currency. A rich ecology has efficient prices for the same reason a rich scientist can afford deep inquiry.

3.3 The reflexive observable

In [11] an observable is a time-varying quantity both parties agree on, and it is exogenous: the weather, a stock price, a block height. Under (i) the observable that sets an engine’s rate reads the engine’s own inventory. This is the one place the combinator language had to be extended, and it is worth naming, because it is what turns a contract into an organism: the contract’s terms depend on the contract’s state.

4 The engine is an organism

4.1 A fourth access profile

[1, §10] separates two ways of being food. A *source* is a persistent emitter: rate-limited, non-exclusive, non-adaptive, inexhaustible on the timescale of one learner though finite in the aggregate. *Prey* is a linear send: exhaustible, exclusive, adaptive. The distinction is not a distinction of sort or of rewrite rule but of multiplicity — a fact about the cut.

The engine is a third way, and it is obtained by taking the same four attributes and choosing a combination the table did not previously contain.

Proposition 4.1 (The engine is derived, not imported). *The profile in Table 1 requires no new sort, no new rewrite rule, and no new GSLT. A persistent receipt on a quote channel supplies persistence and non-exclusivity; two located reserves supply per-colour exhaustibility; and computing*

the rate from those reserves supplies rate-adaptivity. All three are available in the host calculus of [1].

Proof. Appendix A exhibits the term. Persistence is $\leq$; location is a channel, per [8, §2.2]; the rate is a **for**-comprehension over the reserve channels that replaces what it reads. $\square$

Remark 4.2 (The persistence is load-bearing). An engine written with a linear receipt models one conversion, not a converter — the same bug found and fixed in the knot encoding, where crossing gates written with $\leftarrow$ evaporate after a single visitation and model one passage through the diagram rather than the diagram. The correct shape is a *persistent issuer* offering *linear* contracts: the enzyme persists, the substrate binding does not.

Remark 4.3 (What the engine eats). [7, §6] introduces dissipation $\theta(a \rightarrow b) \geq 0$ as a modelling parameter and reads the resulting inequality as the second law. Here θ is not posited. Every rendezvous is metered, and the swap is a rendezvous, so θ is the toll — and it accrues to the engine’s own stack, because something has to fund the engine’s continued existence and nothing else is available. The second law and the engine’s metabolism are the same fact. An engine whose traffic falls below its burn rate starves, exactly as a scientist does.

Remark 4.4 (Not a decomposer). It is tempting to call the engine a decomposer, on the grounds that it lives on what others cannot metabolise. The better analogue is the mycorrhizal network [16]: it does not consume waste, it makes two otherwise incommensurable metabolisms mutually available and takes a cut. That literature is also the right prior art for §6.3, since it studies exchange rates set by partner choice and outside options rather than by no-arbitrage [15].

4.2 The combinators as reaction language

With the organism identified, the combinators of [11] sit one level down: they name the engine’s reactions, not the engine.

one(k) is the funded unit transfer; **give** swaps which signature funds the leg, which [8, §3.1] observes is exactly the two-sided surface of the cut made operational. **scale**(o, c) carries the rate, and by [8, §3.6] scaling multiplies the amount moved and *not* the fuel demand — cost counts funded rendezvous, not magnitude. **and** is the atomic join, under which the partial-funding hazard is structurally impossible.

Observation 4.5 (The swap is the netted futures shape). An engine’s conversion is

$$\mathbf{and}(\mathbf{scale}(o, \mathbf{one}(a)), \mathbf{give}(\mathbf{one}(b))),$$

Pattern (A) of [8, Tab. 1], bilateral cash settlement. That note shows the apparatus rewards the netted form: one transfer in the data-dependent direction instead of two gross legs, atomic by construction and costing one token instead of two.

Corollary 4.6 (Netting is thermodynamic). *Since θ is the toll (Remark 4.3), netting halves the dissipation on every conversion. What [8] presents as a settlement optimisation is, in this reading, a reduction of the second-law loss along every edge of the engine graph, and hence — by §6.4 — an increase in the profitable radius.*

The temporal combinators do work too, and §11 takes them up.

Rung	What C does	What it contributes
relay	forwards faithfully	θ_{toll} only
lossy	drops with probability $1 - p$	θ_{loss} , stranding
corrupting	emits a wrong value	the syndrome of Theorem 6.5
reading	forwards and retains	grade-one access to all traffic
converting	swaps colour against inventory	the engine of Definition 3.2

Table 2: Media classified by behaviour. The engine is the top rung, so the fourth access profile of Table 1 is not a separate kind of thing but the most behaviourally elaborate interposition.

5 The medium

5.1 Interposition is without loss of generality

The host calculus takes communication to be perfect. It does so without loss of generality, because imperfection can be modelled as an additional process rather than an additional rule: races and higher-order channels suffice. Generalise

$$\text{for}(y \leftarrow x)P \mid x!(Q) \quad \text{to} \quad \text{for}(y \leftarrow x_1)P \mid C(x_1, x_2) \mid x_2!(Q),$$

where C is any process joining the two endpoints. The simplest C is a persistent two-way forwarder; richer ones do arbitrarily more.

Proposition 5.1 (The perfect case is a special case). *When C is the persistent two-way forwarder on (x_1, x_2) , the interposed term is weakly bisimilar to the direct one. Hence nothing is lost by working in the general form, and the perfect-communication calculus is recovered as the identity medium.*

Proof sketch. This is the boundary-forwarder lemma of the knot encoding [9], whose persistence is exactly what makes the encoding compositional. The forwarding steps are τ and are absorbed by weak bisimulation; persistence is required, since a linear forwarder relays once and then is not a channel. $\square$

Definition 5.2 (Loss and corruption). A medium *drops* when it receives on x_1 and emits nothing, and *corrupts* when it emits a value other than the one received. Write $p \in (0, 1]$ for the per-hop probability that a message is relayed faithfully.

Remark 5.3 (Interposition is not cost-preserving, and that is θ). Proposition 5.1 is a statement about behaviour, not about metering: a single gated rendezvous becomes at least two. So interposition is not a morphism in the cost-accounted category of [6] — it multiplies the gated transitions rather than preserving them — and the defect is exactly the dissipation θ that §4 attributes to the engine. This is the third instance in the series of one pattern: in [3, §4.3] the lax comparison map’s defect is the interaction; here the interposition’s defect is the cost of distance. Distance is not a parameter of this calculus. Distance is interposition, and its price is the failure of a map to be strict.

5.2 A ladder of media

Remark 5.4 (The intermediary learns most). A reading medium on a bridge has grade-one access to everything crossing it. §10 argues that commerce across an incommensurable boundary can only

be epistemic; Table 2 adds that the boundary itself is an epistemic asset, and that whoever runs the forwarder is best placed to harvest it. This is not a decoration: it predicts that intermediaries accumulate information rents, which is what intermediaries observably do.

5.3 Loss is stranding, and stranding is exergy

Proposition 5.5 (Dropped tokens are stranded, not destroyed). *Under fork (i) the per-colour totals are conserved by construction. A message dropped by a lossy medium is therefore not annihilated: it rests on a channel no process will receive on. Total value ν is conserved; available value strictly decreases.*

Corollary 5.6 (The second law, with a mechanism). *The free-energy reading that [7, Rmk. 6.2] wants and can only posit is here realised. Exergy is ν minus the measure of the stranded set, and it is monotone because stranding is irreversible: recovering a stranded message would require a process that can name the channel it rests on, which by construction is the medium that failed.*

Proposition 5.7 (The two dissipations add). *Along a path of d hops, tolls accumulate additively and reliability compounds as p^d . Writing $\theta = \theta_{\text{toll}} + \theta_{\text{loss}}$ with $\theta_{\text{loss}} = -\log p$, both contributions are additive in the cost metric and the delivered value is $\nu e^{-\theta d}$ as before.*

So Proposition 6.10’s profitable radius is partly an information horizon and not only an economic one: a pathway can price itself out either by charging too much or by losing too much, and the two enter the same exponent.

Remark 5.8 (Corruption is where the syndrome comes from). Theorem 6.5 treats a corrupted rate $a\mathbf{1}_e$ as given. Table 2 says where it comes from: a corrupting medium on the quote path of engine e . The price code is not detecting an abstract misquotation; it is detecting the medium.

5.4 The medium is a namespace, and it is higher order

Since C is a process, $@C$ is a name, and $*@C = C$ recovers the process. Media are therefore nameable, and the media of a region form a namespace.

Definition 5.9 (Medium namespace). For a region B , put

$$\text{Chn}(B) = \{ @C : C \text{ mediates a rendezvous within } B \}.$$

This namespace is higher order in two senses, and both do work. First, C ’s free names are the endpoints it joins, so $@C$ sits one level up the reflection tower from the names it carries: a medium is a name for a carrier of names. Second, a rich C is itself realised by interposition, so media are mediated by media.

Proposition 5.10 (The tower). *Chn at one level is the endpoint namespace at the next: the media over which a region’s parts communicate are the channels of the region taken as a part of something larger. The tower is well-founded downwards, since by the no-self-code theorem a name codes a strictly smaller term, and bounded upwards by Proposition 6.10, since a level whose media exceed the profitable radius has no traffic to mediate.*

Proposition 5.10 is the fractal of §9 stated on the naming side rather than by analogy, and it supplies the generating step that closure under $|$ could not: *the outside of one level is the inside of the next.*

5.5 Inside, outside, and porosity

Individuality is not total closure. An individual that could not be impinged upon could not eat and could not learn. The right notion is partial closure, and the partition it needs is already in the corpus.

Definition 5.11 (Three kinds of medium). Relative to a region B , a medium is *internal* if both endpoints lie in B , *boundary* if exactly one does, and *both-type* if it supports rendezvous of each kind. This is the channel classification of [1, §12.6] applied one level up, to media rather than to endpoints. Write $\text{Chn}_{\text{in}}(B)$, $\text{Chn}_{\partial}(B)$.

Definition 5.12 (Individual). A region B is an *individual* when $\text{Chn}_{\text{in}}(B)$ is closed — every medium of an internal rendezvous is nameable from within B — while $\text{Chn}_{\partial}(B)$ is not. Internal communication may be arbitrarily complex; what is required is not simplicity but ownership.

Proposition 5.13 (Perfect closure is death). *If $\text{Chn}_{\partial}(B) = \emptyset$ then B has no grade-one or grade-two access across its cut. By [1, §5] it cannot harvest, so σ is monotone decreasing and B starves; and it can run no assay, so it cannot revise. Total closure is fatal in both currencies.*

Proposition 5.13 is the individuation-level analogue of the game note’s result that a solved science is a dead ecology. Perfection of a boundary and perfection of a theory fail in the same way.

Proposition 5.14 (Porosity requires redex pathways). *An open boundary name is not yet porosity. For the environment to impinge, a boundary channel must carry a chain of redexes reaching a redex on an internal channel; symmetrically, for the individual to act, an internal redex must reach a boundary one. A boundary channel with no such pathway is a stranding site in the sense of Proposition 5.5: messages arriving there are conserved and unavailable.*

Remark 5.15 (The same undecidability, and the same remedy). “Has a redex pathway to an internal redex” is undecidable in general, for the reason [1, §12.5] gives for “will communicate”. We inherit that note’s remedy: take the syntactic over-approximation — the interaction graph on shared names — and accept that it over-counts.

Definition 5.16 (Integration depth). The *integration depth* of a percept is the length of the redex pathway of Proposition 5.14 from the boundary channel at which it enters to the internal redex at which it is used. Depth one is reflex; greater depth composes information from more of the surface.

Proposition 5.17 (Deep integration is exponentially fragile). *A percept of integration depth d traverses d hops of internal medium and survives with probability p^d . Its cost is d metered rendezvous. Hence for saturating gains $Y(d)$ the net value $Y(d)p^d - cd$ has an interior maximum whose location falls with p .*

Corollary 5.18 (Ownership is what buys depth). *Integration depth is not purchased with budget alone. It is purchased with reliability, and reliability is available only on a medium the individual can name — since retransmission requires grade-two access to $@C$. Definition 5.12 is therefore not a stipulation: closure of the internal medium is the condition under which deep integration is affordable, and deep integration is what distinguishes an individual from a bag of reflexes.*

Computed in Appendix B, block (5), with $Y_0 = 100$, $Y(d) = Y_0(1 - 2^{-d})$, $c = 2$: optimal depth $d^* = 5$ at $p = 0.999$ and $p = 0.99$, falling to 3 at $p = 0.95$ and 0.90 and to 2 at $p = 0.75$, with net value falling from 86.4 to 38.2 across that range. An organism with an unrepaired medium is not merely slower; it is structurally shallower.

Remark 5.19 (Two axes). Table 2 classifies media by *behaviour*; Definition 5.11 classifies them by *ownership*. Individuation lives on the second axis only. An engine may be internal or boundary, and which it is decides whether the region containing it is one economy or two; what it does decides how much it costs to be either.

Remark 5.20 (Individuation is a fixed point). Definition 5.12 partitions $\text{Chn}(B)$ relative to a choice of B , and B was to be determined by the criterion. This is a fixed-point condition rather than a construction, and its multiple solutions are the levels of §9: the circularity is the fractality. We state the condition and do not claim an algorithm for finding its solutions.

6 The engine graph

6.1 Two graphs, one geometry

There are now two graphs in play and they must not be confused. The *composition graph* has learners as vertices and shared names as edges; its edge data is the grading vector $\mathbf{k}$, and [3, §5] reads ecological relations off it. The *engine graph* Γ has colours as vertices and engines as edges; its edge data is the log-rate ℓ and the dissipation θ .

They are related but not identical. An engine is a shared name, hence a spanning redex, hence an edge of the composition graph; and since the shared name is one at which stacks are held and drawn, it is of type **Met**. So:

Proposition 6.1 (Conversion is metabolic overlap). *Two learners linked by an engine have $k_{\text{met}} \geq 1$. Hence by the factorisation theorem of [3, §4.2] they are not probabilistically independent, and by Proposition 7.1 below their overlap cannot be positive-sum in tokens.*

Remark 6.2 (The vertices are the atomic colours). By [7, Prop. 4.4] a valuation is a homomorphism out of Σ and induces no engine; in particular no engine converts a compound authority $s_1 * s_2$ into a single s' , since that would launder the multi-signature guarantee. Compound signatures are therefore priced but unreachable: they are vertices of no engine. Throughout, Γ is taken on the atomic colours, and β_1 is computed there.

Definition 6.3 (The cost metric). Weight each edge of Γ by its dissipation θ and let d_θ be the induced path metric. This is the geometry the composition note lacked, and it is the only geometry available: the framework has no ambient space, so “far” can only mean *expensive*, in the sense of the geometric weight of [6].

6.2 The cycle space is a code

Recall from [7] that ℓ is arbitrage-free iff it is exact, $\ell = -\delta p$, and that $C^1 = \text{im } \delta \oplus \ker \delta^*$ splits orthogonally into the exact (gradient) and cycle parts, with $\dim \ker \delta^* = \beta_1(\Gamma)$. Write Π_{Cut} and Π_{Cyc} for the two projections, and for an engine e let $R_{\text{eff}}(e)$ be its effective resistance in Γ with unit conductances. Recall the standard identity $R_{\text{eff}}(e) = (\Pi_{\text{Cut}})_{ee}$.

Definition 6.4 (Syndrome). For an observed log-rate cochain ℓ , the *syndrome* is $\ell_{\text{arb}} = \Pi_{\text{Cyc}} \ell$. It is zero iff the observed rates are consistent with some price list.

Theorem 6.5 (Detection strength). *Let the true rates be exact and let a single engine e be corrupted by a in log-rate, $\ell = \ell_{\text{true}} + a \mathbf{1}_e$. Then*

$$\|\ell_{\text{arb}}\|^2 = a^2(1 - R_{\text{eff}}(e)).$$

In particular the corruption is undetectable iff $R_{\text{eff}}(e) = 1$ iff e is a bridge of Γ .

Proof. ℓ_{true} is exact so $\Pi_{\text{Cyc}}\ell_{\text{true}} = 0$ and $\ell_{\text{arb}} = a \Pi_{\text{Cyc}}\mathbf{1}_e$. Since Π_{Cyc} is an orthogonal projection, $\|\Pi_{\text{Cyc}}\mathbf{1}_e\|^2 = (\Pi_{\text{Cyc}})_{ee} = 1 - (\Pi_{\text{Cut}})_{ee} = 1 - R_{\text{eff}}(e)$. An edge lies in no cycle iff it is a bridge, and $\mathbf{1}_e \perp \ker \delta^*$ iff $\Pi_{\text{Cyc}}\mathbf{1}_e = 0$; equivalently $R_{\text{eff}}(e) = 1$, which for unit conductances characterises bridges. $\square$

Theorem 6.6 (The code’s total strength is β_1). *For any connected engine graph,*

$$\sum_{e \in E} (1 - R_{\text{eff}}(e)) = |E| - (|V| - 1) = \beta_1(\Gamma).$$

Proof. Foster’s theorem [12] gives $\sum_e R_{\text{eff}}(e) = |V| - 1$; equivalently $\text{tr } \Pi_{\text{Cut}} = \text{rank } \delta = |V| - 1$. Subtract from $|E| = \text{tr } I$. $\square$

Theorem 6.6 is the sharp form of the note’s second claim. A price system carries a fixed budget of self-checking, equal to the number of independent arbitrage cycles, and the geometry of the engine graph decides how that budget is distributed across engines. Dense regions get most of it; bridges get none. Verified numerically in Appendix B: for the barbell of two K_5 ’s joined by a single engine, $\beta_1 = 12$ and $\sum_e (1 - R_{\text{eff}}) = 12.000000$, with the bridge contributing 0 and each cluster edge contributing 0.6.

Proposition 6.7 (Series engines are confusable). *Two corrupted engines e, f produce the same syndrome direction iff $\Pi_{\text{Cyc}}\mathbf{1}_e \parallel \Pi_{\text{Cyc}}\mathbf{1}_f$, which holds iff e and f lie in exactly the same cycles. Hence engines in series — a chain through vertices of degree two — cannot be told apart by the code, while engines in parallel can.*

In C_6 , where $\beta_1 = 1$, all 15 edge pairs are confusable: a one-dimensional cycle space detects that *something* is wrong and never what. In the barbell, no pair is. The code localises exactly when the ecology is dense.

Corollary 6.8 (Density buys detection). *In a d -regular engine graph on n colours the mean detection strength is exactly $1 - 2(n - 1)/(nd)$, approaching $1 - 2/d$. Measured: 0.344 at $d = 3$, 0.672 at $d = 6$, 0.902 at $d = 20$ for $n = 60$.*

6.3 Bridges carry no pricing discipline

Theorem 6.5 has a consequence that reframes the interstellar picture. A thin pathway between two dense components is a cut edge; cut edges lie in no cycle; so no-arbitrage constrains a bridge rate *not at all*. Path-independence is vacuous when there is one path.

Corollary 6.9 (The numeraire is global but uninformative across a cut). *If Γ is connected then a valuation ν exists globally and is unique up to gauge, by [7, Thm. 3.3]. But the rate on a bridge is unconstrained by the no-arbitrage condition: any value of it extends to a consistent ν . Within a dense component each rate is over-determined by many independent cycles; across a bridge it is under-determined.*

So the right statement is about *redundancy*, not existence. This also inverts the compression proposition of [7, §5]: the collapse from $O(n^2)$ bilateral rates to $O(n)$ prices is exactly the redundancy, read as a code. Compression and error correction are the same β_1 counted from opposite ends, which is precisely the trade-off [4] is about.

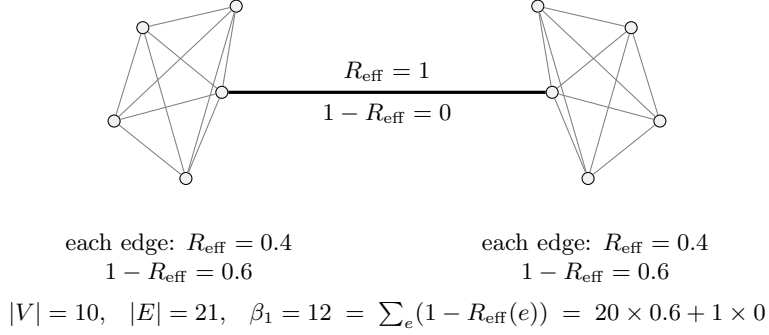


Figure 1: The price code on a barbell engine graph. Twenty intra-cluster engines each carry 0.6 of detection strength; the bridge carries none. The total is β_1 exactly (Theorem 6.6). A misquoted rate inside a cluster produces a syndrome and is localisable; a misquoted rate on the bridge is invisible.

6.4 Decay along a pathway

By [7, Prop. 6.1], dissipation makes ν non-increasing along engine operations. Along a path of length d in the cost metric the delivered value is $\nu e^{-\theta d}$. Set against the foraging inequality's requirement that a harvest exceed the cost of obtaining it, and taking a round-trip toll linear in d , a conversion at distance d is profitable iff $Y e^{-\theta d} > c d$.

Proposition 6.10 (Profitable radius). *For $Y, c > 0$ and $\theta > 0$ the profitable set is an interval $[0, d^*)$ with d^* the unique root of $Y e^{-\theta d} = c d$. There is a horizon on conversion set by metering, not by any speed limit.*

Computed in Appendix B for $Y = 1000$, $c = 20$: $d^* = 19.17$ at $\theta = 0.05$, 8.73 at $\theta = 0.2$, 1.52 at $\theta = \ln 10$, 0.82 at $\theta = 5$.

Remark 6.11 (The ecological pyramid is θ). Predation is an engine: it converts prey-colour into predator-colour, and it is spectacularly lossy. The classical figure of roughly ten per cent transfer efficiency per trophic level [14] is $r = 1/10$, i.e. $\theta = \ln 10 = 2.303$ per engine. The energy pyramid is then exactly the exponential decay of ν along a path of engines, which is Proposition 6.1 of [7] with no adjustment. We did not arrange this and record it as a check rather than a result.

7 What conversion does to composition

We audit the six results of [3] under the addition of engines.

Factorisation survives and is sharpened. An engine is a spanning redex, so composed learners linked by one are dependent (Proposition 6.1). Two invariants of the same graph now coexist: $\mathbf{k}$ counts edges of dependence, β_1 counts cycles of it.

No metabolic mutualism needs qualification. The proposition of [3, §6.1] says that when overlap is confined to Met and Src, $\Delta_1 + \Delta_2 \leq 0$. Two things must be said.

Proposition 7.1 (Dead stock). *Under (i) the total token supply of each colour is conserved by every engine operation, and by [7, Prop. 5.1] the total ν is conserved in the frictionless limit and*

strictly decreases under θ . So conversion is never positive-sum in value. But residual life is stack divided by burn rate, and a token whose signature a learner cannot spend contributes zero to its residual life and non-zero to the other's. Hence conversion can strictly increase $\sum_i \sigma_i/b_i$ while conserving ν .

Remark 7.2 (An honest gap in the source). The proof sketch in [3, §6.1] argues that a transfer over a shared metabolic name is “strictly negative in aggregate residual life whenever the two burn rates differ”. That step is directional: with $\sum_i \sigma_i/b_i$ as the aggregate, a transfer toward the slower burner raises it. In one colour the asymmetry is easy to miss because the transfers that raise the aggregate are exactly the ones a predator has no reason to make. Colours make it conspicuous, because dead stock is worth zero to its holder by construction, so the profitable direction and the aggregate-raising direction coincide. We therefore read the composition note’s proposition as correct about ν and in need of the qualification above about residual life, and we take the qualification to be the content of “gains from trade” in this setting.

Corollary 7.3 (The slogan, amended). Cooperation is epistemic; competition is metabolic — except where the metabolisms are incommensurable, in which case conversion is cooperative in residual life and still competitive in value. The positive-sum region is exactly the region where signatures do not match.

Competitive dilution is re-coupled. [3, §9] shows that learners sharing a rate-limited source take shares $\propto 1/\text{cost}$ and that the dilution is regressive. Engines reinstate that competition across boundaries that rooting had made clean: an engine is a shared name, so conversion turns effective $k_{\text{src}} = 0$ into effective competition.

Corollary 7.4 (Markets sell back modularity). Rooting-by-quotation makes disjointness a theorem and thereby purchases modularity; an engine sells it back at the price of the toll. Modularity and tradability are opposed, and the grading vector prices the trade.

Phase, merger, succession. Phase remains non-compositional. Merger is now sharply distinguished from composition-with-engines: merger identifies namespaces, an engine prices them, and §8 says when each is available. Succession — a change of the grading vector forced by source exhaustion — acquires a second mechanism: an engine’s reserve running out changes Γ ’s topology, and by Theorem 6.6 the code’s total strength changes with it.

Corollary 7.5 (Extinction degrades the price system). Removing an engine from a cycle reduces β_1 by one and redistributes R_{eff} ; removing the last engine on a cut disconnects Γ and destroys the common numeraire. Loss of a converter is loss of commensurability, not merely of throughput.

8 Individuation: where a namespace closes

8.1 The three parameters, found inside the framework

[4] locates the boundary between a shared substrate and autonomous replication at a critical radius $r^* \sim \sqrt{\rho v/\varepsilon}$. To transpose it we must find ε , ρ , and v inside the present framework rather than import them.

- ε , the **error rate**, is the per-hop drop-or-corrupt rate $1 - p$ of the interposed medium (§5). This is the channel [4] is parameterised on — error rate on transmission — and interposition supplies it natively rather than by analogy.

- ρ , the **repair throughput**, is the rate of retransmission, which by Corollary 5.18 is available only where the medium is nameable.
- v , the **communication speed**, is $1/\theta$. There is no speed in a cost-accounted calculus, but there is the temporal stack, and by [6, Prop. 4.1] time is the count of forced redexes. An expensive pathway simply *is* a slow one, and by Proposition 5.7 it is expensive for two reasons that enter the same exponent.

Accidental name collision, which an earlier draft of this section took as the error channel, is a second and different one: its failure mode is loss of disjointness rather than loss of a message, and its repair is re-rooting by freshness-by-quotation rather than retransmission. [3, §7] proves it occurs. Hypothesis error is a third. §8.6 returns to the question of which binds.

Remark 8.1 (Why the medium channel is the one that individuates). Collision threatens the *identity* of names and is repaired by re-quoting, which a region can do unilaterally. Loss threatens the *delivery* of messages and is repaired by retransmission, which a region can do only over media it owns. Since Definition 5.12 turns on ownership of the internal medium, it is the transmission channel that fixes the individuation threshold, and collision that fixes how finely a namespace may be subdivided within it.

8.2 The structural form: partial closure

Proposition 8.4 below gives a quantitative threshold. Definition 5.12 gives a structural one — the internal medium is closed, the boundary medium is not — and the two are the same condition seen from two sides.

Proposition 8.2 (Closure is what makes repair possible). *Retransmission over a medium C requires grade-two access to $@C$, so ρ is supported only on $\text{Chn}_{\text{in}}(B)$. A region whose internal media are not all nameable from within has ρ effectively reduced on the unnameable part, and by Proposition 8.4 its sustainable radius falls accordingly.*

Corollary 8.3 (Bridges fail twice). *A boundary medium belongs to no region's closed internal namespace, so neither party can unilaterally repair it. Together with Theorem 6.5 — a bridge lies on no cycle and so carries no syndrome — a thin pathway is uncorrectable for two independent reasons: nothing checks its rates, and nobody owns its medium.*

8.3 Isoperimetry replaces dimension

The derivation in [4] balances errors accumulating over a volume r^d against repair delivered through a cross-section r^{d-1} , with a propagation factor v/r . On a graph there is no dimension; the analogues are the ball $B(r)$ and its edge boundary $\partial B(r)$, and their ratio is the isoperimetric ratio $h(B) = |\partial B|/|B|$.

Proposition 8.4 (Individuation threshold). *A region B of the engine graph may close over its namespace — may be merged, in the sense of [3, §8] — iff repair keeps pace with collision, that is iff*

$$\rho |\partial B| \frac{1}{\theta r} \geq \varepsilon |B|, \quad \text{equivalently} \quad h(B) \geq \frac{\varepsilon \theta}{\rho} r.$$

Writing $\kappa = \varepsilon \theta / \rho$, the individual is the largest ball satisfying $h(B(r)) \geq \kappa r$.

Corollary 8.5 (Recovering r^*). *On a graph of polynomial growth d one has $h(B(r)) \sim c/r$, so the condition becomes $c/r \geq \kappa r$ and $r^* = \sqrt{c/\kappa} = \sqrt{c\rho/\varepsilon\theta}$, which is the formula of [4] with $v = 1/\theta$.*

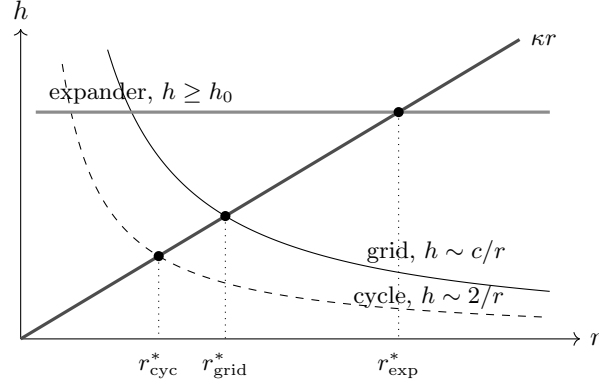


Figure 2: Individuation as a crossing. A region may be merged while its isoperimetric ratio lies above the line κr , $\kappa = \varepsilon\theta/\rho$. Polynomially growing graphs cross early and give $r^* \propto \kappa^{-1/2}$; an expander holds h up and crosses late, giving $r^* \propto \kappa^{-1}$ and, since balls grow exponentially in r , an individual exponentially larger in population.

8.4 Expansion

Theorem 8.6 (Density buys exponentially larger individuals). *If Γ restricted to B is an expander, $h(B(r)) \geq h_0 > 0$ uniformly, then $r^* = h_0/\kappa$ is linear rather than square-root in $\rho/\varepsilon\theta$; and since an expander has diameter $O(\log |B|)$, the merged population satisfies $|B(r^*)| = e^{\Omega(r^*)}$. On a graph of polynomial growth d the merged population is only $|B(r^*)| = O((r^*)^d)$.*

Proof. The first claim is Proposition 8.4 with h bounded below. The second is the standard fact that a family with $h \geq h_0$ has diameter $O(h_0^{-1} \log n)$, so balls grow exponentially in radius. $\square$

Theorem 8.6 is the point at which the three threads of this note become one. The “densely connected component” of the motivating picture, the region where the price code is strong, and the region that may close over its namespace are all the same region, and expansion is the common criterion. Effective resistance is small exactly where expansion is good; by Theorem 6.5 detection is strong exactly where effective resistance is small; and by Proposition 8.4 merger is sustainable exactly where the boundary keeps up with the bulk.

Appendix B, block (2), computes the merged population at four values of κ for a cycle, a 20×20 grid, and random 3- and 6-regular graphs on 400 vertices. At $\kappa = 0.10$ the cycle sustains an individual of 5 colours, the grid 85, the 3-regular graph 160, and the 6-regular graph 365 — the whole population.

8.5 What merger deletes

The intuition that centralisation buys compression and loses error correction is right, and the framework can say precisely what is lost, because merger destroys two different things.

Proposition 8.7 (Merger sets $\beta_1 = 0$). *A merged region has one colour, hence no engines internal to it, hence no cycles in Γ restricted to it. By Theorem 6.6 its total detection strength is zero. Centralisation does not degrade the price code; it deletes it.*

Proposition 8.8 (Merger destroys independence). *By the factorisation theorem of [3, §4.2], disjointness is equivalent to $\mu_{12} = \mu_1 \otimes \mu_2$. Merger identifies namespaces, so the joint distribution of a merged region does not factor and failures are correlated.*

These are detection and containment respectively, and they attach to ε and ρ respectively. It is worth noting that the second is exactly the third bullet of [4, §1] — the Fisher information matrix is block-diagonal in the autonomous case and densely off-diagonal in the shared case — restated in the composition note’s own vocabulary. The two notes agreed before either knew it.

Against these, merger buys: no toll on internal transfers, since there are no engines to pay; no arbitrage, since there are no cycles to carry it; and the coordination compression of [7, §5] in its limiting form, where one does not even need the n prices. That is the trade, priced on both sides in one currency.

8.6 Which ε binds

[4] asks which error channel is rate-limiting in a cell and conjectures proteostasis rather than genome integrity. We inherit the question with three candidates, each with its own repair process and its own throughput: *transmission loss*, repaired by retransmission and used above; *name collision*, repaired by re-rooting; and *hypothesis error* — the degrading prediction that the motivation mechanism of [1, §2] punishes — repaired by revision. They may well bind at different levels of the hierarchy, and the levels are where the question becomes sharp: an organism’s internal medium is repairable and a pod’s is not, so transmission should bind higher up and collision lower down. The composition rule proposed in [4], $r^* = \min_i r_i^*$, applies; which i attains the minimum at each rung is open.

9 The hierarchy

9.1 Closure gives nesting; r^* gives levels

Composition of learners yields a learner, so the construction is closed under an associative operator and self-similarity costs nothing. But closure permits arbitrary nesting; it does not select levels.

Proposition 9.1 (Levels are forced by a finite r^*). *If r^* is finite, then a population exceeding $|B(r^*)|$ cannot be maintained as one closed namespace, and the only way to continue scaling is by replication of r^* -sized units composed with engines. Each composite has its own effective $\varepsilon, \rho, \theta$, hence its own r^* . The hierarchy is a sequence of critical radii.*

The generating step is Proposition 5.10: the outside of one level is the inside of the next. An orca’s boundary media — calls, touch, the water — are the pod’s internal media, and the pod is an individual exactly to the extent that it can name and repair them. So the tower of media and the tower of individuals are the same tower, and neither has to be posited alongside the other.

This is also the argument of [4, §6] connecting r^* to West’s allometry [17]: the invariance of the terminal unit is a consequence rather than a premise, and fractal replication above it is forced. Here the same argument produces the levels the composition note could only permit. The sequence terminates where $\theta \rightarrow \infty$ and even the cheapest channel prices out.

9.2 The shutoff ordering

Which components of $\mathbf{k}$ survive a given pathway cost? Price the *maintenance* of each type of overlap.

Observation 9.2 (Ordering by maintenance cost). *Met pooling requires continuous rendezvous and is the most expensive to maintain. Mat requires periodic rendezvous. Src requires co-availability at a rate-limited emitter. Prb requires only that a legible surface stand where both*

Boundary	Operator	Surviving overlap	Relation
within a skill set	$\bowtie$	all	one budget, one namespace
organism / organism	+ engine	Met (transfer), Src, Prb, Mat	kin, trade, culture
deme / deme	+ engine	Src, Prb, Mat	competition, culture
species / species	+ engine	Src, Prb	predation, eavesdropping
biosphere / biosphere	only	Prb	signals
beyond d^*	—	none	nothing

Table 3: The hierarchy as a shutoff sequence. Each row is a level boundary, identified by which component of the grading vector has gone to zero and whether merger is still available. The ordering is Observation 9.2; the availability of $\bowtie$ is Proposition 8.4.

can read it, and by the read-versus-drive distinction of [2, §5.1] a written surface costs nothing to leave standing. Hence as θ rises the components extinguish in the order Met, Mat, Src, Prb.

9.3 The grading vector is stratified, and has a fifth component

Proposition 9.3 (A fifth typed namespace). *Chn is a typed namespace in the sense of [3, Def. 4.1], and overlap in it is not reducible to overlap in the other four. Two learners sharing a medium can observe each other’s traffic and can congest each other, relations that Met, Src, Prb, Mat cannot express: eavesdropping, jamming, and contention for bandwidth. Hence*

$$\mathbf{k} = (k_{\text{met}}, k_{\text{src}}, k_{\text{prb}}, k_{\text{mat}}, k_{\text{chn}}).$$

Remark 9.4 (This is very likely the arena). [3] leaves open whether the vector is complete and names the *arena* as the candidate fifth type; [2] realises the arena as the process routing moves between players. A router is a medium. We therefore conjecture that the arena namespace and Chn are the same namespace reached from two directions. We record this here rather than amending [3], so that the two notes do not disagree in print before that note is revised.

Corollary 9.5 (Composition and merger are the two halves of k_{chn}). *Sharing a boundary medium is composition: the learners talk, interfere, and can trade. Sharing an internal medium is merger, by Definition 5.12. So the fifth component’s in/out split is exactly the distinction [3, §8] draws between its two operators, which is evidence that the split is a refinement rather than an epicycle.*

Corollary 9.6 ($\mathbf{k}$ is a sequence). *By Proposition 5.10 the grading vector is relative to a stratum, so the object that describes a composite fully is a sequence of vectors, one per level of the medium tower. Table 3 reads off the sequence for one lineage.*

Table 3 has a pleasing consequence and a testable one. The pleasing one: the last surviving channel is Prb, which is exactly the channel [3, Obs. 6.1] identifies as the only positive-sum one, because formulae are not conserved. The testable one: the depth of the hierarchy is the length of the grading vector, so the open question of whether there is a fifth typed namespace — the arena — is the question of whether there is a fifth level.

9.4 An orca

An orca must solve hunting, position in the pod, mating, and care of young. Suppose each is undergirded by a learner in the sense of [3]. Then:

The skills share one budget, so within the organism there is one colour, no engines, and $\beta_1 = 0$: *there is no market inside an individual*. This is Coase’s theory of the firm [18] arriving unbidden.

Two claims must be kept apart here, and an earlier draft of this note ran them together. That there is no *market* inside an individual is a claim about colour change: no rung-five medium, no conversion, allocation by hierarchy rather than by price. That there is no *mediation* inside an individual would be a claim about the lower rungs of Table 2, and it is false — an orca’s internal communication is as elaborate as anything it does, and by Corollary 5.18 that elaboration is precisely what closure of the internal medium buys. Mediation is not allocation. The boundary of the individual is where the token colour changes *and* where the medium stops being owned; the two coincide because an unowned medium cannot sustain a shared budget.

The pod is not merged. Provisioning and food-sharing are transfers at rates, hence composition with engines, and the framework predicts what is observed: reciprocity with terms, not pooling.

Competitive dilution becomes a theory of skill acquisition. Skills at a shared rate-limited source take shares $\propto 1/\text{cost}$, and the regressive corollary of [3, §9] says dilution costs the unfinished learner its depth and the finished one nothing. Inside one organism this says a mature skill crowds out a developing one *without any contact between them*, which is automatisisation and the critical period, with a closed form.

And the instability of non-interference becomes development. Skills inside one organism are rooted at the same quotation, so name coincidence is not accidental but generic: the individual is precisely the regime in which the disjointness theorem fails by construction. That is why skills interfere, and why they transfer.

Remark 9.7 (The orca claim is empirical, not definitional). “Each skill is backed by a learner” has content only if the skill learners have distinct populations. If they are merely different formulae held by one population, this is one learner with a large relaxation lattice and the composition apparatus adds nothing. The discriminator is [1, Prop. 12]: a single learner cannot leave its ball. So if skills let each other escape basins, they must be separate populations, and transfer between skills is the test.

10 Incommensurability

Life elsewhere may not be organised around the same chemistry, and in this framework that is not a matter of expense but of topology.

Proposition 10.1 (No engine, no numeraire). *If no engine links the colours of two learners then Γ is disconnected, the connectedness hypothesis of [7, Thm. 3.3] fails, and there is no common valuation: each component carries its own ν with its own gauge. Incommensurability is disconnection.*

Theorem 10.2 (Commerce across an incommensurable boundary is epistemic). *Let $\mathcal{L}_1, \mathcal{L}_2$ have disjoint Met and Src namespaces and no engine between their colours. Then no token can pass between them. But by [3, Obs. 6.1] a formula is not conserved: a quotation @P is inert while held, transmissible, and inspectable at grade three, and one learner’s coming to hold it does not deprive the other. Hence the only available commerce is over Prb.*

Theorem 10.2 is the slogan of [3, §6] scaled up one level: between biospheres, cooperation is not merely the cheapest positive-sum channel but the *only* channel. Signals cross; food does not. That is also consistent with Observation 9.2, since Prb is the last component to extinguish, and the two arguments are independent.

Proposition 10.3 (First contact is a coincidence, not a construction). *An engine requires a rendezvous, hence a shared name. Two ecologies rooted at independent quotations are disjoint by [3, §7]; but by the same section’s remark, independently manufactured names can coincide accidentally, and the exposure lemma of [1, §9] already records that closure is luck rather than construction. So the establishment of a first engine between ecologies is a structural coincidence whose probability is bounded by the name-growth constraint of the no-self-code theorem.*

We stop here. The obvious next question — what a population of expanding, converting ecologies does under these constraints, and whether Proposition 6.10’s horizon reproduces or refutes the grabby aliens hypothesis [19] — is left to §14.

11 Credit, default, and the discount rate

The temporal combinators `truncate( $t, c$ )` and `get( $c$ )` denote claims on future value. In an ecology with a fixed Θ this looks impossible: credit smells like minting. It is not.

Proposition 11.1 (Credit is not minting). *A forward is a located slot whose release is gated on a temporal condition. Nothing is created: the tokens exist and are held, and what transfers is the right to receive on the slot. This is exactly the funding slot of [8, §3.2], with the guard on the temporal stack rather than on a signature.*

Proposition 11.2 (The discount rate is the hazard rate). *A borrower that fails to repay in this setting is a starved borrower: by [10, Def. 1] and the metabolic law of [1, §5], failure to fund is death. Credit risk is therefore mortality risk exactly, and the rate at which a lender discounts a future claim is the borrower’s hazard rate, which the framework computes from stack over burn rate.*

Remark 11.3 (Options price the exploration budget). The choice combinators come with the conservative-bound discipline of [8, §3.5]: reserve the maximum over branches, refund the unforced remainder. That is the exact structure of the question [1, §6] leaves open — how much should a learner pay to keep an unresolved hypothesis open? An option on a revision move has a premium, a strike in metered units, and a refund on lapse, and the search-strategy catalogue can be re-read as a portfolio problem. We flag this and do not pursue it.

12 On deriving the rates

[7] takes the rate function r as given, and so has this note. It should be derivable. The natural candidate is that the rate between two colours is the ratio of marginal inquiry-yield per metered unit, which [2] computes exactly for its ladders: the non-modal rungs of the noughts ladder deliver +0.444 of total gain for 87.9 of 255.9 units, and the Nim ladders of [3] give returns that are flat in one environment and rising in another.

If that identification holds, then ν is a field over the ladder rather than a bare price list, and the composition note’s headline — whether truth is affordable is a property of the environment, not the learner — becomes a statement about exchange rates: communities working environments where truth is cheap should systematically price their tokens differently from communities where it is dear. This is checkable inside the existing simulations and we regard it as the most tractable of the open problems.

13 What is shown, and what is not

Shown, in the sense of proved from the imported apparatus: Theorems 6.5, 6.6 and 8.6; Propositions 3.3, 6.1, 6.7, 8.4, 6.10; and the merger propositions of §8. These are graph theory and linear algebra over a structure the earlier notes construct; they inherit whatever the earlier notes' constructions are worth.

Shown by computation: the numbers of Appendix B, which are exact where rational arithmetic was used and float otherwise.

Not shown: that the rates are what §12 conjectures; which of the three error channels of §8.6 binds at which level; that Chn is [3]'s arena, which is conjectured and not proved; that the identification of v with $1/\theta$ is the only defensible one, as opposed to the most natural; that inventory engines are the right reading of the conservation fork, as against (ii); that the isoperimetric balance in Proposition 8.4 has the right propagation factor, which is inherited from [4] and is heuristic there. The orca discussion is illustration, not evidence, and §9 flags the test that would make it evidence.

Not attempted: coherence of the lax monoidal structure of [3]; any dynamics on Γ beyond the quasi-static drift of Proposition 3.3; any treatment of strategic behaviour by engines, which is where biological market theory [15] would enter.

14 Open problems

1. **Which ε binds, and at which level.** Name collision and hypothesis error are two channels with two repair processes. Make $r^* = \min_i r_i^*$ precise and determine the minimising i at each rung of Table 3.
2. **Derive the rates.** Execute §12 against the existing ladders and test whether exchange rates track environmental structure as predicted.
3. **Dynamics on the engine graph.** Under (i) the marking drifts, so β_1 is fixed but R_{eff} moves and engines can die. Run the composite of [3, §10] with engines under a Gillespie scheme and ask whether the ecology self-organises toward an expander.
4. **Is Chn the arena?** Proposition 9.3 adds a fifth typed namespace and conjectures it is [3]'s arena. Settling this requires revising that note's relations table with a fifth column and checking that the entries — eavesdropping, jamming, contention — are not already derivable from the other four.
5. **Reordering.** Proposition 5.5 covers dropping and Table 2 covers corruption, but a medium may also reorder, which is a race rather than a fault and which stranding does not describe. Whether reordering is a third dissipation or a change in the observed transition system is open.
6. **A syntactic criterion for porosity.** Proposition 5.14 needs a decidable test that over-approximates the existence of a redex pathway from a boundary channel to an internal redex. Reuse the interaction graph of [1, §12.5] and measure the over-count.
7. **The assay trichotomy under loss.** A lossy medium introduces a second source of the $\perp$ outcome that has nothing to do with budget, and the learner cannot tell the two apart from inside. Budget-relative refutation is therefore not well defined without a hypothesis about

the channel — so a learner must do science on its own medium. This is a correction owed to [1, §4] rather than a problem for this note, and we flag it here because this note is what raises it.

8. **Merger as a quotient, arbitrage-removal as a reflection.** $\bowtie$ identifies colours; Π projects rates. Both take a conversion structure to a simpler one, and [7, §4] shows the second is an idempotent reflection. Is there an adjunction relating them, with Proposition 8.4 as the condition under which the left adjoint exists?
9. **OSLF factoring and the crossover.** [4] conjectures that r^* and the factoring of the generated logic over a population are two views of one threshold. Proposition 8.4 gives the first a graph-theoretic form; give the second one and compare.
10. **Grabby aliens.** With Proposition 6.10's horizon, Theorem 10.2's restriction to signals, and the coincidence bound on first contact, the expansion of a converting ecology is bounded by profitability rather than by light speed. Whether that reproduces, softens, or refutes the selection effects of [19] is a question this framework can now pose.

A An engine, in rholang

The listing below gives located reserves, the reflexive observable, the acceptance gate, the combinators, and the engine itself as a persistent issuer offering a linear swap. Conventions follow [2]: a parameter written `c` is a name and `@d` is data; a channel is passed as `*c`, since $@(*c) = c$; persistent receipts use `<=`. Guards marked `///!` *ideal* use the `where` clause in a form the current transpiler does not accept.

```
// engine.rho -- an inventory engine, its reflexive observable, and the
// SPJ combinators that name its reactions. Companion to "Engines and
// Individuals". Conventions follow the game note: a parameter written
// 'c' is a NAME and '@d' is DATA; a channel is passed as '*c', since
// @(*c) = c. Persistent receipts use '<='; a linear receipt is '<-'.
// Guards marked '///!' ideal use the where-clause of rholang 1.4 in a
// form the current transpiler does not yet accept.

new Engine, Observable, Reserve, Reserves,
  One, Give, Scale, And, Swap,
  Gate, Spend, Trade, Wall, Boot in {

  // -----
  // 1. Located reserves.
  //
  // A reserve is a stack held ON A CHANNEL, never floating in the soup.
  // The right to draw is the right to receive on that channel. Two
  // reserves, one per colour, constitute an engine's inventory.
  // -----

  contract Reserve(@lbl, @sig, @n, ret) = {
    new slot in {
      slot!((sig, n)) | ret!(*slot)
    }
  } |

  // -----
  // 2. The reflexive observable.
  //
  // In Peyton-Jones's combinators an observable is an exogenous
  // quantity both parties agree on. Here it reads the issuer's own
  // inventory: the rate IS a function of the marking. This is the one
```



```

// place the combinator language had to be extended.
// -----

contract Observable(slotA, slotB, ret) = {
  for (@(sa, na) <= *slotA & @(sb, nb) <= *slotB) {
    // constant-product quote:  $r(a \rightarrow b) = R_b / R_a$ 
    slotA!((sa, na)) | slotB!((sb, nb)) | ret!(nb * 1.0 / na)
  }
} |

// -----
// 3. The acceptance gate.
//
// Peel k tokens of a given signature off a located slot, all or
// nothing. Either a committed private purse comes back, or the slot
// is left exactly as it was. This is the operational content of the
// linear resource proof: supply at least demand, decided BEFORE any
// leg fires.
// -----

contract Gate(slot, @sig, @k, ret) = {
  for (@(s, n) <- *slot) {
    match (s == sig) and (n >= k) {
      true => {
        slot!((s, n - k)) |
        new purse in { purse!((sig, k)) | ret!((true, *purse)) }
      }
      false => { slot!((s, n)) | ret!((false, "insufficient")) }
    }
  }
} |

contract Spend(purse, @k, ret) = {
  for (@(s, n) <- *purse) {
    match n >= k {
      true => { purse!((s, n - k)) | ret!((true, s)) }
      false => { purse!((s, n)) | ret!((false, Nil)) }
    }
  }
} |

// -----
// 4. The combinators.
//
// A contract is a channel that, sent an acquisition message,
// discharges what it denotes. 'one' moves a unit and is funded by
// the payer; 'give' swaps which side funds; 'scale' multiplies the
// amount by an observable, NOT the fuel demand -- cost counts funded
// rendezvous, not magnitude; 'and' is the atomic join.
// -----

contract One(@colour, src, dst, purse, ret) = {
  new ok in {
    Spend!(*purse, 1, *ok) |
    for (@(fired, _) <- ok) {
      match fired {
        true => for (@(c, n) <- *src) {
          src!((c, n - 1)) |
          for (@(d, m) <- *dst) { dst!((d, m + 1)) | ret!(true) }
        }
        false => ret!(false)
      }
    }
  }
} |

contract Give(@colour, src, dst, purse, ret) = {

```

```

    // the two-sided surface of the cut, made operational: swapping
    // which signature funds the leg is the whole of 'give'.
    One!(colour, *dst, *src, *purse, *ret)
} |

contract Scale(obs, @base, ret) = {
  new q in { obs!(*q) | for (@r <- q) { ret!(base * r) } }
} |

contract And(c1, c2, purse, ret) = {
  // reserve the JOINT demand first, then fire both legs under one
  // gate. Half-settlement is structurally impossible inside a join.
  new g, r1, r2 in {
    Gate!(*purse, "fuel", 2, *g) |
    for (@(okg, joint) <- g) {
      match okg {
        true => { c1!(joint, *r1) | c2!(joint, *r2) |
                  for (@a <- r1 & @b <- r2) { ret!(a and b) } }
        false => ret!(false)
      }
    }
  }
} |

// -----
// 5. The engine.
//
// A PERSISTENT issuer standing over two located reserves, offering a
// linear swap contract on demand. The persistence is not cosmetic:
// a linear receipt here would model one conversion, not a converter.
// The enzyme persists; the substrate binding is linear.
//
// The swap is the netted futures shape of the finance note:
// and( scale(o, one(a)), give(one(b)) )
// performed as ONE transfer in the data-dependent direction, which
// costs one token instead of two and cannot half-settle.
// -----

contract Engine(slotA, slotB, tollSlot, quote, swap) = {
  // the quote face: anyone may read the rate. Perception is grade
  // one and must not be able to act, so 'quote' is a different name
  // from 'swap'.
  contract quote(ret) = { Observable!(*slotA, *slotB, *ret) } |

  // the trade face: a linear acquisition, funded by the taker, with
  // the toll accruing to the engine's own stack. The engine eats.
  contract swap(@dir, @amt, takerPurse, ret) = {
    new g, q in {
      Gate!(*takerPurse, "fuel", 1, *g) |
      for (@(okg, purse) <- g) {
        match okg {
          true => {
            quote!(*q) |
            for (@r <- q) {
              match dir {
                "ab" => {
                  for (@(sa, na) <- *slotA & @(sb, nb) <- *slotB) {
                    // theta is not a parameter: it is this rendezvous.
                    slotA!((sa, na + amt)) |
                    slotB!((sb, nb - amt * r)) |
                    for (@(t, k) <- *tollSlot) { tollSlot!((t, k + 1)) } |
                    ret!((true, amt * r))
                  }
                }
              }
            }
          }
          _ => {
            for (@(sa, na) <- *slotA & @(sb, nb) <- *slotB) {
              slotA!((sa, na - amt * 1.0 / r)) |
            }
          }
        }
      }
    }
  }
}

```

```

        slotB!((sb, nb + amt)) |
        for (@(t, k) <- *tollSlot) { tollSlot!((t, k + 1)) } |
        ret!((true, amt * 1.0 / r))
    }
}
}
}
}
false => ret!((false, Nil))
}
}
}
}
} |

// -----
// 6. A wall, two communities, one engine.
//
// The wall is the shared rate-limited source of the composition note:
// persistent, non-exclusive, non-adaptive, finite. The engine is the
// fourth access profile: persistent and non-exclusive like a source,
// exhaustible per colour like prey, and adaptive in its rate --- the
// only one of the four whose terms respond to who is trading.
// -----

contract Wall(@remaining, out) = {
    match remaining > 0 {
        true => { out!(1) | Wall!(remaining - 1, *out) }
        false => Nil // heat death, on schedule
    }
} |

contract Trade(swap, @dir, @amt, purse, ret) = {
    //! ideal: only trade when the quote leaves us solvent
    //! for (@r <= quote where r * amt > minAcceptable) { ... }
    swap!(dir, amt, *purse, *ret)
} |

contract Boot(ret) = {
    new rA, rB, rT, q, sw, w in {
        Reserve!("engine-a", "a", 100, *rA) |
        Reserve!("engine-b", "b", 100, *rB) |
        Reserve!("engine-toll", "fuel", 0, *rT) |
        for (@slotA <- rA & @slotB <- rB & @slotT <- rT) {
            Engine!(@slotA, @slotB, @slotT, *q, *sw) |
            Wall!(12000, *w) |
            ret!((*q, *sw))
        }
    }
} |

Boot!(*Reserves)
}

// -----
// 7. Media.
//
// Rho's communication is perfect, and that is without loss of
// generality: a pathway is a PROCESS interposed between the endpoints.
// for (y <- x)P | x!(Q)
// generalises to
// for (y <- x1)P | C(x1, x2) | x2!(Q)
// and C may do arbitrarily more than relay. The rungs below are the
// behaviour axis of the ladder; whether a given C's quote lies inside a
// region's namespace is the ownership axis, and it is the second that
// individuates.
// -----

```

```

new Relay, Lossy, Corrupting, Reading, Chain in {

  // rung 1: the identity medium. PERSISTENT -- a linear forwarder
  // relays once and is then not a channel at all.
  contract Relay(in, out) = {
    for (@m <= *in) { out!(m) }
  } |

  // rung 2: drops with probability 1 - p. A dropped message is not
  // destroyed; it is left resting on a channel nothing receives on.
  // Conservation survives; availability does not. That gap is exergy.
  contract Lossy(in, out, @p, coin) = {
    new strand in {
      for (@m <= *in) {
        new r in {
          coin!(*r) |
          for (@u <- r) {
            match u < p {
              true => out!(m)
              false => strand!(m) //! stranded, not annihilated
            }
          }
        }
      }
    }
  } |

  // rung 3: emits a value other than the one received. On a quote path
  // this is exactly the single-edge fault the price code detects.
  contract Corrupting(in, out, noise) = {
    for (@m <= *in & @d <= *noise) { out!(m + d) }
  } |

  // rung 4: forwards and retains. Grade-one access to all traffic --
  // the intermediary learns most, and on a bridge it learns everything.
  contract Reading(in, out, log) = {
    for (@m <= *in) { out!(m) | log!(m) }
  } |

  // rung 5 is Engine, above: a medium that swaps colour against its own
  // inventory. The ladder is one construction throughout.

  // a pathway of length d, with reliability p^d and toll d. This is
  // what "distance" means here; there is no other geometry.
  contract Chain(in, out, @d, @p, coin) = {
    match d {
      0 => Relay!(*in, *out)
      _ => new mid in { Lossy!(*in, *mid, p, *coin) |
                        Chain!(*mid, *out, d - 1, p, *coin) }
    }
  }
}

```

Three things to watch, tying the code to the propositions. The **quote** face and the **swap** face are different names, because perception must not be able to act — grade one and grade two separated by choice of namespace, as in [2, §4.3]. The toll accrues to the engine’s own slot on every swap, which is Remark 4.3: θ is not a parameter but a rendezvous. And **Observable** reads the reserves and replaces them, which is Proposition 3.3: the rate is a function of the marking, so the term itself refuses to let arbitrage be a static property.

B Reproducing the numbers

All figures in this note come from `engines.py`, four blocks.

(1) The price code. Barbell of two K_5 's joined by one engine: $|V| = 10$, $|E| = 21$, $\beta_1 = 12$. An exact cochain has syndrome 2.3×10^{-15} . Corrupting a single edge by $a = 1$ gives $\|\ell_{\text{arb}}\|^2 = 0.6$ for every intra-cluster edge ($R_{\text{eff}} = 0.4$) and 0 for the bridge ($R_{\text{eff}} = 1$); the identity of Theorem 6.5 holds to 1.8×10^{-15} . The Foster identity of Theorem 6.6 is verified at $12.000000 = \beta_1$ for the barbell, and separately for K_5 (6), C_6 (1), the 5×5 grid (16) and a random 3-regular graph on 60 vertices (31). Confusable pairs: 0 in the barbell, all 15 in C_6 . Mean detection strength for random d -regular graphs on 60 vertices: 0.344, 0.508, 0.672, 0.803, 0.902 at $d = 3, 4, 6, 10, 20$, against the $1 - 2/d$ asymptote.

(2) Individuation. Isoperimetric profiles for C_{400} , the 20×20 grid, and random 3- and 6-regular graphs on 400 vertices, and the largest ball satisfying $h(B(r)) \geq \kappa r$:

κ	C_{400}	grid	3-regular	6-regular
0.30	3	25	22	151
0.10	5	85	160	365
0.03	11	219	258	365
0.01	19	315	349	365

(3) Decay. $Y = 1000$, $c = 20$, d^* solving $Ye^{-\theta d} = cd$: 19.172 at $\theta = 0.05$; 8.728 at 0.20; 1.518 at $\ln 10$; 0.822 at 5.0.

(4) Inventory engines. A triangle of constant-product engines, each with reserves (100, 100), has cycle yield exactly 1. Draining one engine in steps of ten carries the yield through

$$0.827, 0.683, 0.562, 0.457, 0.367,$$

that is, $\log \oint \ell$ from 0 to -1.0033 . Computed in exact rationals.

(5) Porosity. With $Y_0 = 100$, $Y(d) = Y_0(1 - 2^{-d})$, $c = 2$, and net value $Y(d)p^d - cd$, the optimal integration depth and its net value are:

p	d^*	net	p^{d^*}
0.999	5	86.392	0.9950
0.990	5	82.127	0.9510
0.950	3	69.020	0.8574
0.900	3	57.788	0.7290
0.750	2	38.188	0.5625

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